

ServiceNow Architecture Showcase: End-to-End Telecom ITSM Solution

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Executive Summary

The Telecom Incident Management application is a fully scoped, B2B ITSM solution engineered for Telecom Providers. Designed with multi-tenant readiness (Domain Separation) in mind, this application streamlines the reporting, routing, and resolution of critical network outages. It bridges the gap between external customer systems, self-service portals, and internal IT fulfillment teams using OOTB best practices, secure APIs, and automated workflows.

1. Foundation & Multi-Tenant Architecture

The core of the application was built to ensure data isolation, scalability, and seamless deployment across environments.

- **Scoped Application & Source Control:** Bootstrapped in a dedicated scope (x_1520874_teleco_0) with strict GitFlow integration for CI/CD readiness.
- **Domain Separation Readiness:** Engineered to support a Provider/Customer data model (e.g., Telecom Provider -> Bank ABC), utilizing company-driven domain routing Business Rules to automatically isolate tenant data without heavy GlideRecord queries.
- **Custom Data Model:** Built the Telecom Incident table extending the native Task table, featuring custom auto-numbering (TINC) and specialized dictionary fields (Network Technology, Outage Scale, Estimated Restoration Time).

Domain map:

fetch Configuration Item (CI) details, preventing UI freezing and adhering to modern ServiceNow performance standards.

- **Immutable Audit Trails:** Implemented a strict *Before Delete* Business Rule and Data Policies to prevent the unauthorized deletion of critical outages (National/City-level) and enforce mandatory resolution documentation.

dynamic Info Message appearing when a CI is selected:

The screenshot shows a ServiceNow incident form for 'Telecom Incident TINC1042'. At the top, a blue bar displays 'Network Element Info | Location: 13308 Midland Road, Poway, CA | Category: Hardware'. The form fields include: Number (TINC1042), Priority (1 - Critical), * Company (City Hospital), State (Open), Assignment group (Service Desk), * Estimated Restoration Time (2026-08-15 02:19:39), Assigned to, * Affected Network Element (*RON-IBM), Outage Scale (National Outage), and Network Technology (-- None --). An 'Active' checkbox is checked.

Prevent users from deleting critical outage records (National/City-level):

The screenshot shows a ServiceNow incident form for 'Telecom Incident TINC1043'. A red banner at the top states: 'Security Alert: Deletion of National or City-level outage records is strictly prohibited by company compliance policy.' The form fields include: Number (TINC1043), Priority (1 - Critical), * Company (City Hospital), State (Open), Assignment group (Service Desk), Estimated Restoration Time (2026-08-22 01:58:06), Assigned to, Affected Network Element (*ANNIE-IBM), Outage Scale (Street Level), and Network Technology (5G). An 'Active' checkbox is checked. Below the form, there is a 'Short description' field with 'City Hospital Alert' and a 'Description' field containing a 'Raw Alert Details' block with the text: 'URGENT NETWORK ALERT FROM CITY HOSPITAL', 'Hello Support, We are facing a critical outage in our main database cluster. Server Name: SRV-DB-01. Please resolve this immediately.'

3. Robust Integrations (REST & Email)

To support enterprise B2B customers, the system exposes secure channels for automated incident creation.

- **Scripted REST API:** Developed a strict JSON Inbound API endpoint for "Bank ABC" with basic authentication, granular role access, and robust payload validation (guard clauses) returning HTTP 400 for missing mandatory fields.
- **Inbound Email Parsing:** Engineered a bullet-proof Inbound Email Action for "City Hospital" that utilizes Regex HTML stripping and zero-query domain routing (via System Properties) to translate legacy email alerts into actionable P1 incidents.

Scripted REST API:

POST https://dev279471.servi: + ...

https://dev279471.service-now.com/api/x_1520874_teleco_0/bank_abc_integration/incident

POST https://dev279471.service-now.com/api/x_1520874_teleco_0/bank_abc_integration/incident

Params Authorization Headers (12) Body Pre-request Script Tests Settings Cookies Beautify

none form-data x-www-form-urlencoded raw binary JSON

```
1 {
2   "network_node": "ROUTER-NY-11",
3   "priority": "2",
4   "short_description": "Connection timeout to main gateway test",
5   "outage_scale": "1",
6   "network_technology": "3",
7   "affected_network_element": "27d415a8c0a8000b00ffe2ab0f82e8d2"
8 }
```

Body Cookies (4) Headers (20) Test Results 201 Created 1977 ms 1.32 KB Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "result": {
3     "status": "success",
4     "incident_number": "TINC1067",
5     "message": "Telecom incident successfully created securely for Bank ABC."
6   }
7 }
```

Telecom Incident TINC1067

Number

* Company

Assignment group

Assigned to

Outage Scale

Active ☒

* Short description

Description

Additional comments (Customer visible)

Priority

State

Estimated Restoration Time

Affected Network Element

Network Technology

☐ Work notes

Inbound Email:

4. Flow Designer & SLA Automation

- **Domain-Agnostic Flow Designer:** Architected a global Master Flow for P1 incidents featuring data-driven lookups (via Group Types) for approvals. Engineered advanced trigger gating to completely eliminate PlanRecursionException loops.
- **SLA Engine:** Engineered a 2-hour Resolution SLA for Critical incidents. Refined the Stop/Cancel conditions to gracefully handle skipped/cancelled tickets, preventing skewed performance metrics and "Ghost Data".

Flow Designer:

Telecom Incident
TINC1069

Number

TINC1069

* Company

Bank ABC

Assignment group

Assigned to

Outage Scale

-- None --

Active

☒

* Short description

Test flow

Priority

1 - Critical

State

Open

Estimated Restoration Time

Affected Network Element

Network Technology

-- None --

Discuss

Follow

U

EXECUTION DETAILS

Critical Telecom Incident Automation

Waiting

Cancel flow

Open flow

Open context record

Show Action Details		State	Start time	
1	Look Up Record	Core Action	Completed	2026-08-13 03:33:10 23ms
2	If If empty	Flow Logic	Evaluated - False	2026-08-13 03:33:10 0ms
3	Update Record	Core Action	Not Run	
4	End	Flow Logic	Not Run	
5	Ask For Approval	Core Action	Waiting	2026-08-13 03:33:10 1218ms
6	If If Approved	Flow Logic	Not Run	
7	Update Record	Core Action	Not Run	
8	Create Record	Core Action	Not Run	

EXECUTION DETAILS

Critical Telecom Incident Automation

Completed

Open flow

Open

Show Action Details		State	Start time
3	Update Record	Core Action	Not Run
4	End	Flow Logic	Not Run
5	Ask For Approval	Core Action	Completed 2026-08-13 03:33:10
6	If If Approved	Flow Logic	Evaluated - True 2026-08-13 03:38:18
7	Update Record	Core Action	Completed 2026-08-13 03:38:18
8	Create Record	Core Action	Completed 2026-08-13 03:38:18

when Approved

Telecom Incident
TINC1069

Number

TINC1069

* Company

Bank ABC

Q

Q

Assignment group

Q

Assigned to

Q

Outage Scale

-- None --

Priority

1 - Critical

State

Work in Progress

Estimated Restoration Time

Q

Affected Network Element

Q

Network Technology

-- None --

Active

☒

* Short description

Test flow

Q

Q

Description

Additional comments (Customer visible)

Additional comments (Customer visible)

Q

Work notes

Post

Telecom Incident
TINC1069

Activities: 3

System

Critical approval granted. Maintenance task dispatched.

Work notes • 2026-08-13 03:38:18

System

State Work in Progress was Open

Field changes • 2026-08-13 03:38:18

HS Haroon Saleh

Impact 3 - Low
Opened by Haroon Saleh
Priority 1 - Critical
State Open

Field changes • 2026-08-13 03:33:08

Update

Delete

Related Links

Show SLA Timeline

Repair SLAs

[SN Utils] Versions (0)

Approvers (6)

Tasks (2)

Task SLAs (1)

Number

Search

Parent = TINC1069

	Number	Priority	State	Assigned to	Short description	Tas
☐	GAPRV0010063	4 - Low	Closed Complete	(empty)		Gro
☐	TASK0020810	4 - Low	Open	(empty)	Field Maintenance: Test flow	Tas

SLA Engine:

Approvers (6)

Tasks (2)

Task SLAs (1)

SLA definition

Search

Task = TINC1069

	SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
☐	Telecom P1 Resolution SLA	SLA	Resolution	In progress	1 Hour 54 Minutes	5 Minutes	4.44%	2026-08-13 03:33:08	(empty)

1 to 1 of 1

Technical Stack & Skills Demonstrated

Category	Technologies & Concepts Applied
Backend Automation	Flow Designer, Business Rules, Script Includes, SLAs
Frontend & UX	Service Portal, Record Producers, Client Scripts, UI Policies
Integrations	Scripted REST APIs (JSON), Inbound Email Actions (Regex)
Architecture & Security	Domain Separation, ACLs, User Criteria, Git/Source Control